Homework 15

Please reread all of Section 3.4 of Durrett (on joint distributions).

Please also read Section 5.1 and Section 5.1.1 in Durrett through the end of Example 5.5. (You are welcome to continue to Example 5.6 as well.) You do not need to memorize Durrett's calculations with integrals. Seeing integration by parts in Example 5.5 will help you remember how that technique of integration works. The most important things to take away from the examples, though, are the formulas for expected value and variance for standard continuous distributions.

1. Please work Exercise 43 from **Chapter 3**. (This question is about golf, in honor of Mac and Nathan.) The textbook answer is 56/65. Show the work to calculate this probability using the Bayes' formula method.

   Let A be the event that the golfer made par, and let B be the event that he hit is drive in the fairway. We are given P(A | B) = 0.8, P(A | B^c) = 0.3, and P(B) = 0.7. We calculate P(B | A) using the Bayes' formula method.

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(B)P(A|B)}{P(A)}.$$

   Using the given information, we find $P(B)P(A|B) = (0.7)(0.8) = 0.56$. We use the law of total probability to calculate $P(A)$:

$$\begin{aligned} P(A) &= P(B)P(A|B) + P(B^c)P(A|B^c) \\ &= (0.7)(0.8) + (0.3)(0.3) = 0.56 + 0.09 = 0.65. \end{aligned}$$

   Therefore, $P(B|A) = \frac{0.56}{0.65} = \frac{56}{65}$.

2. The following question is based on Durrett's Exercise 62. Random variables X and Y have the joint distribution given in the following table.

Y	X = 1	2	3	marginal dist. of Y
1	0.1	0.2	0.3	0.6
2	0.15	0.15	0	0.3
3	0.05	0	0.05	0.1
marginal dist. of X	0.3	0.35	0.35	

a. Calculate the marginal distributions, which are the separate distributions for X and Y. I added a blank column and a blank row to the table where you would put the marginals.

b. Are the random variables X and Y independent? Answer using your marginal distributions and the numbers in the joint-distribution table.
   The random variables are not independent. It is enough to find one probability in the joint distribution that is not the product of the corresponding marginals. For example, we have $P\big((X = 2) \cap (Y = 3)\big) = 0$ but $P(X = 2)P(Y = 3) = (0.35)(0.1) = 0.035$.

c. Calculate P(X=1|Y=1) and P(X=2|Y=2).

$$P(X=1|Y=1)=\frac{P\big((X=1)\cap(Y=1)\big)}{P(Y=1)}=\frac{0.1}{0.6}=\frac{1}{6}$$

$$P(X=2|Y=2)=\frac{P\big((X=2)\cap(Y=2)\big)}{P(Y=2)}=\frac{0.15}{0.3}=\frac{1}{2}$$

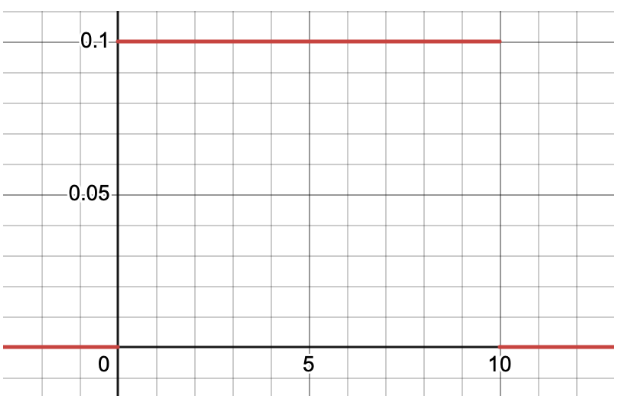
3. Suppose that X and Y are independent random variables. The joint-distribution table below is blank except for the marginal distributions of X and Y. Use the marginal distributions to fill in the table.

Y	X = 0	2	3	marginal dist. of Y
1	$(0.1)(0.5)=0.05$	$(0.2)(0.5)=0.1$	$(0.7)(0.5)=0.35$	0.5
5	$(0.1)(0.3)=0.03$	$(0.2)(0.3)=0.06$	$(0.7)(0.3)=0.21$	0.3
7	$(0.2)(0.1)=0.02$	$(0.2)(0.2)=0.04$	$(0.7)(0.2)=0.14$	0.2
marginal dist. of X	0.1	0.2	0.7	

4. Imagine picking a real number between 0 and 10 so that (roughly speaking) every number is equally likely. Your number is a random variable X, which we describe using the density function f defined by $f(x)=0$, if $x<0$; $f(x)=\frac{1}{10}$, if $0\leq x\leq 10$; $f(x)=0$, if $x>10$.

a. Sketch a graph of this density function.

The following Desmos link shows a graph of the density function: https://www.desmos.com/calculator/m9coyw6hwx.

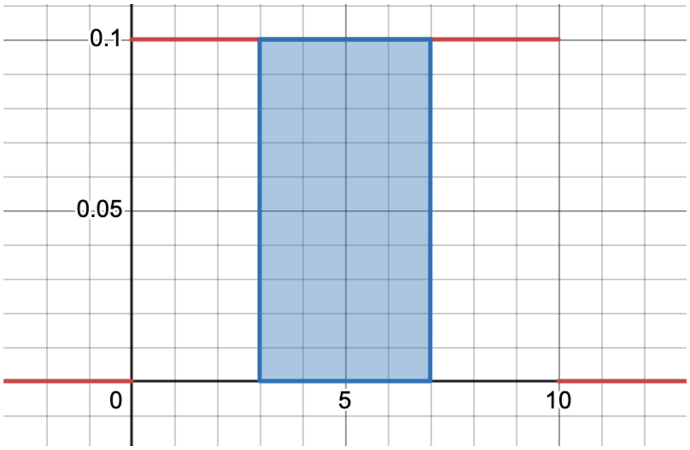


b. Calculate $P(3\leq X\leq 7)$. Shade the part of your graph corresponding to the event $(3\leq X\leq 7)$.

To calculate the probability, we integrate:

$$\int_3^7 f(x)dx=\int_3^7\frac{1}{10}dx=\left[\frac{x}{10}\right]_{x=3}^{x=7}=\frac{7}{10}-\frac{3}{10}=\frac{4}{10}.$$

The following Desmos link shows the shaded graph: https://www.desmos.com/calculator/tya8nwujpo



c. The expected value of X can be calculated as $\int_0^{10} xf(x)dx$. This integral looks like the sum ($\sum_x xf(x)$, where x runs over the values of X) that we use to calculate expected value of a discrete random variable. For a continuous random variable, we replace $f(x)$ with $f(x)dx$ and the sum with an integral. Calculate $E[X]$.

$$E[X] = \int_0^{10} xf(x)dx = \int_0^{10} \frac{x}{10}dx = \left[\frac{x^2}{20}\right]_{x=0}^{x=10} = 5$$

d. The Law of the Unconscious Statistician also works for continuous random variables. Calculate $E[X^2]$ (second moment of X) as $\int_0^{10} x^2 f(x)dx$. Then calculate $\mathrm{Var}(X)$ as $E[X^2] - E[X]^2$.

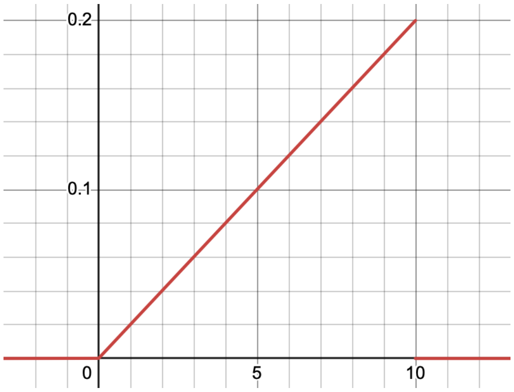
$$E[X^2] = \int_0^{10} x^2 f(x)dx = \int_0^{10} \frac{x^2}{10}dx = \left[\frac{x^3}{30}\right]_{x=0}^{x=10} = \frac{100}{3}$$

$$\mathrm{Var}(X) = E[X^2] - E[X]^2 = \frac{100}{3} - 25 = \frac{25}{3}.$$

5. Imagine you pick two numbers independently at random from the uniform distribution on [0, 10]. That is the distribution from the previous question where, roughly speaking, every number in [0, 10] is an equally likely pick. Let Y be the higher of the two numbers that you pick. The density function f for Y has the following formula: $f(x) = 0$, if $x < 0$; $f(x) = x/50$, if $0 \leq x \leq 10$; $f(x) = 0$, if $x > 10$.

a. Sketch a graph of this density function.

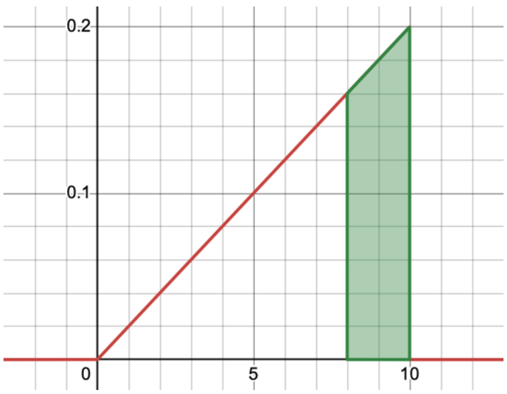
The following Desmos link has a graph of the density function: https://www.desmos.com/calculator/l8yiciu6ep.



b. Use this density function to calculate $P(8 \le Y \le 10)$, the probability that the higher of the two numbers is at least 8. Shade the part of your graph corresponding to the event $(8 \le Y \le 10)$.

$$P(8 \le Y \le 10) = \int_8^{10} \frac{x}{50}\,dx = \left[\frac{x^2}{100}\right]_{x=8}^{x=10} = \frac{100}{100} - \frac{64}{100} = \frac{36}{100}.$$

The following Desmos link has the shaded graph: https://www.desmos.com/calculator/pkeaimmktl.



c. Calculate $E[Y]$.

$$E[Y] = \int_0^{10} x\frac{x}{50}\,dx = \int_0^{10} \frac{x^2}{50}\,dx = \left[\frac{x^3}{150}\right]_{x=0}^{x=10} = \frac{100}{15} = \frac{20}{3}$$

d. Calculate $E[Y^2]$ (the second moment) and $\mathrm{Var}(Y)$.

$$E[Y^2] = \int_0^{10} x^2 \cdot \frac{x}{50}\,dx = \int_0^{10} \frac{x^3}{50}\,dx = \left[\frac{x^4}{200}\right]_{x=0}^{x=10} = 50$$

$$\mathrm{Var}(Y) = E[Y^2] - E[Y]^2 = 50 - \left(\frac{20}{3}\right)^2 = 50 - \frac{400}{9} = \frac{50}{9}$$